Appendix E. Evaluation Record

This appendix gives the record behind the comparisons of Section 4: every pairwise comparison (Table E.20); every recourse evaluation those comparisons use, with its stopping status, band, gap and slack (Table E.21); the neighbours of the plan without storage re-evaluated under the certification rule (Table E.22); the solver starts of the static arrangements (Table E.23); and the evaluations refused by the gap clause (Table E.24). It closes with the certification statistics and the reproducibility record. The multi-scenario evaluations, stopped by the production exit rather than certified, are reported in Table 13. All entries are generated from the archived evaluation records. Throughout, Q is the gross operational cost, settlement excluded; the net form subtracts the terminal salvage credit (38); the value of a plan is $Q(\mathbf{0}) - Q(\boldsymbol{x})$ under the same flexibility price; $F = I + Q$; and $\tau = 4{,}539.07$ €. Plans are written node: converter rating (MVA) / energy capacity (MWh). Amounts are rounded to 0.1 k€ and ratios to two decimals.

Appendix E.1. Comparisons

Table E.20 lists the 58 distinct comparisons of the record. Fifty-five are reported both gross and net of the salvage credit; the remaining three, the reference unit at 1.75 times the base flexibility price and the two comparisons at the 0.50 end-of-life floor, were added later and are reported gross only. Each comparison is decided by the rule of Appendix A.5: between two certified evaluations it is determinate if $|\Delta| \geq \max\{3b, 2\tau\}$, with b the larger band; with an uncertified evaluation, if $|\Delta|$ exceeds three times the largest consensus gap and settling slack involved, in gross and gap-corrected terms. Two certificates rest on a turning point at a cycle that was not clean (evaluations 4 and 38); the comparisons that use them are also given in the uncertified form in the note to the table, and none changes verdict.

Table E.20: Every pairwise comparison of the record. Eval.: the two evaluations of Table E.21 (reference / other); ^{u} uncertified; ^{f} flagged certificate. Thr.: the determinacy threshold, or the uncertified bar where an uncertified evaluation is involved; ×: $|\Delta|$ divided by it. det.: determinate; w.r.: within resolution; w.b.: within the uncertified bar. Net: net of the terminal salvage credit, against the same threshold or bar. Gross operational cost, settlement excluded.

			Gross				Net of salvage		
No.	Comparison	Eval.	Δ (k€)	Thr. (k€)	×	Verdict	Δ (k€)	×	Verdict
(a) Plans against the plan without storage at the baseline price: $\Delta = F(\boldsymbol{x}) - F(\mathbf{0})$; positive: the plan without storage costs less									
1	7: 0.25/1 (reference unit)	1 / 2	64.4	13.1	4.93	det.	64.4	4.93	det.
2	7: 0.5/2	1 / 3	135.5	13.4	10.13	det.	135.5	10.13	det.
3	7: 0.75/3	1 / 4^{f}	208.3	12.6	16.49	det.	208.3	16.49	det.
4	7: 1/4	1 / 5	274.0	13.6	20.19	det.	274.0	20.19	det.
5	7: 1.25/5	1 / 6^{u}	347.2	23.9	14.52	det.	347.2	14.52	det.
6	7: 0.5/1	1 / 7	114.1	12.6	9.03	det.	114.1	9.03	det.
7	7: 1/2	1 / 8^{u}	231.7	24.2	9.56	det.	231.7	9.56	det.
8	7: 1.5/3	1 / 9^{u}	361.7	41.4	8.74	det.	361.7	8.74	det.
9	7: 2/4	1 / 10	494.4	13.4	36.91	det.	494.4	36.91	det.
10	7: 2.5/5	1 / 11	616.6	13.4	45.94	det.	616.6	45.94	det.
11	5: 0.25/1	1 / 12	58.0	13.2	4.39	det.	58.0	4.39	det.
12	9: 0.25/1	1 / 13	61.1	12.6	4.84	det.	61.1	4.84	det.
13	9: 0.75/3	1 / 14	199.1	13.5	14.78	det.	199.1	14.78	det.
14	5: 0.25/0.5	1 / 15	49.7	12.6	3.94	det.	49.7	3.94	det.
15	5, 9: 0.25/0.5 each	1 / 22	102.1	12.6	8.09	det.	102.1	8.09	det.
16	5, 7: 0.25/0.5 each, 2030	1 / 23	93.9	13.4	7.00	det.	84.0	6.27	det.
17	7: 0.5/1, 2030	1 / 26	99.0	12.8	7.73	det.	88.3	6.89	det.
18	7: 0.5/1, 2035	1 / 27	139.7	12.6	11.06	det.	85.1	6.74	det.
19	7: 0.5/2, 2030	1 / 28	137.4	12.6	10.88	det.	113.1	8.95	det.
20	7: 0.5/2, 2035	1 / 29	223.6	12.6	17.71	det.	109.1	8.64	det.
(b) Flexibility price m, plans in 2025 against the plan without storage at the same price: $\Delta = \text{value} - I$; positive: the plan pays									
21	7: 0.25/1, $m = 1.5$	30 / 31^{u}	−14.5	18.8	0.77	w.b.	−14.5	0.77	w.b.
22	7: 0.25/1, $m = 1.75$	32 / 33	14.2	10.8	1.32	det.	—	—	—
23	7: 0.25/1, $m = 2$	34 / 35	46.3	12.3	3.77	det.	46.3	3.77	det.
24	7: 0.5/2, $m = 2$	34 / 36	92.8	12.8	7.26	det.	92.8	7.26	det.
25	7: 0.75/3, $m = 2$	34 / 37^{u}	133.8	27.7	4.83	det.	133.8	4.83	det.
26	7: 1/4, $m = 2$	34 / 38^{f}	176.6	9.7	18.14	det.	176.6	18.14	det.
27	7: 1.25/5, $m = 2$	34 / 39^{u}	228.4	26.5	8.60	det.	228.4	8.60	det.
28	7: 0.25/1 → 0.5/2, $m = 2^{\Delta}$	35 / 36	46.5	12.8	3.64	det.	46.5	3.64	det.
29	7: 0.5/2 → 0.75/3, $m = 2^{\Delta}$	36 / 37^{u}	41.0	27.7	1.48	det.	41.0	1.48	det.
30	7: 0.75/3 → 1/4, $m = 2^{\Delta}$	37^{u} / 38^{f}	42.8	27.7	1.55	det.	42.8	1.55	det.
31	7: 1/4 → 1.25/5, $m = 2^{\Delta}$	38^{f} / 39^{u}	51.8	26.5	1.95	det.	51.8	1.95	det.
(c) Twice the base flexibility price, 2030 unless stated: neighbours against the plan returned by the search (5: 0.25/0.5; 7: 1/3.5, evaluation 40), $\Delta = F(\text{neighbour}) - F(\text{returned plan})$; positive: the returned plan costs less									
32	7: 0.75/3, 2025	40^{u} / 37^{u}	94.3	27.7	3.41	det.	133.4	4.81	det.
33	5: 0.25/0.5; 7: 0.75/3	40^{u} / 43	28.6	27.7	1.03	det.	32.4	1.17	det.
34	5: 0.25/0.5; 7: 1/3	40^{u} / 42^{u}	26.4	27.9	0.95	w.b.	34.1	1.22	det.
35	5: 0.25/0.5; 7: 1.25/3	40^{u} / 44	53.3	27.7	1.92	det.	62.8	2.27	det.
36	5: 0.25/1; 7: 1/3	40^{u} / 41^{u}	−6.1	27.9	0.22	w.b.	−5.4	0.19	w.b.

Table E.20 (continued)

No.	Comparison	Eval.	Δ (k€)	Thr. (k€)	×	Verdict	Δ (k€)	×	Verdict
37	5: 0.25/0.5; 7: 0.75/3; 9: 0.25/0.5	40^{u} / 45^{u}	9.8	28.2	0.35	w.b.	11.0	0.39	w.b.
38	7: 0.75/3	40^{u} / 47	61.5	27.7	2.22	det.	69.2	2.50	det.
39	7: 1/3	40^{u} / 46	50.6	27.7	1.83	det.	61.1	2.21	det.
40	7: 1/3.5	40^{u} / 48	23.9	27.7	0.86	w.b.	26.8	0.97	w.b.
41	7: 1/4	40^{u} / 49	6.4	27.7	0.23	w.b.	1.6	0.06	w.b.
42	7: 0.75/3; 9: 0.25/0.5	40^{u} / 52	35.3	27.7	1.28	det.	38.5	1.39	det.
43	7: 1/3; 9: 0.25/0.5	40^{u} / 50^{u}	31.3	27.7	1.13	det.	39.2	1.41	det.
44	7: 1/3.5; 9: 0.25/0.5	40^{u} / 51^{u}	5.6	27.7	0.20	w.b.	5.8	0.21	w.b.
45	5: 0.25/0.5; 7: 0.75/2.5^{n}	40^{u} / 53	47.7	27.7	1.72	det.	58.5	2.11	det.
	(d) Ageing calibrations of the reference unit (7: 0.25/1, 2025) at the baseline price: Δ = value − I (positive: the unit pays), or the difference of the two values (positive: the first calibration gives the higher value)								
46	No ageing	1 / 54	−4.1	12.9	0.32	w.r.	−4.1	0.32	w.r.
47	Cycling only	1 / 55	−31.6	12.6	2.50	det.	−31.6	2.50	det.
48	Baseline^{t}	1 / 56	−64.4	13.1	4.93	det.	−64.4	4.93	det.
49	Datasheet 8 000 cycles	1 / 57	−45.2	12.6	3.58	det.	−45.2	3.58	det.
50	Retention 50 %	1 / 58	−73.7	12.8	5.74	det.	−73.7	5.74	det.
51	Retention 50 %, mid-block	1 / 59	−65.9	12.9	5.12	det.	−65.9	5.12	det.
52	Baseline, 0.50 floor	1 / 60	−59.5	12.6	4.71	det.	—	—	—
53	No ageing − ret. 50 %	58 / 54	69.6	12.9	5.41	det.	69.6	5.41	det.
54	Cycling only − ret. 50 %	58 / 55	42.1	12.8	3.28	det.	42.1	3.28	det.
55	Baseline − ret. 50 %	58 / 56	9.3	13.1	0.71	w.r.	9.3	0.71	w.r.
56	Datasheet − ret. 50 %	58 / 57	28.5	12.8	2.22	det.	28.5	2.22	det.
57	Ret. 50 %, mid-block − ret. 50 %	58 / 59	7.9	12.9	0.61	w.r.	7.9	0.61	w.r.
58	Baseline: 0.50 − 0.70 floor	2 / 60	4.9	13.1	0.38	w.r.	—	—	—

Note: Row 1 is also the headline comparison of Subsection 4.1; rows 17–20 are each recorded once gross and once net. In the uncertified form, row 3 has a bar of 70.7 k€ (2.94×) and row 26 a bar of 15.9 k€ (11.12×, the form used in Table 10), both determinate; rows 30 and 31 already use the uncertified form. $^{\Delta}$: the increase of value $-I$ from the smaller to the larger plan (positive: the added capacity pays). Ret. 50 %: the 50 % retention calibration. ^{n}: not a unit neighbour of the returned plan (its energy capacity at node 7 differs by two lattice steps). ^{t}: evaluation 56 reproduces evaluation 2 bitwise, so row 48 repeats row 1 with the sign of value $-I$. Row 58 against evaluation 61, which reproduces evaluation 2 bitwise through cycle 172, gives the same figures. —: not evaluated net.

Appendix E.2. Evaluations

Table E.21 lists the 61 recourse evaluations behind Table E.20 and the tables of Section 4, grouped by the comparisons they serve. For a certified evaluation it gives the certification cycle k^{*}, the range of the objective over the certifying window as a fraction of τ and the band b that enters the determinacy threshold; for an uncertified one, the cycle at which it stopped, the cause, and the priced interface-consensus gap and the settling slack that enter its bar. The last three columns give the number of cycles after the first residual pass with a local solve that was not clean (excluded from certification evidence), the cycle through which a repeated evaluation replayed its

earlier record bitwise, and whether the rule was applied during the run or to the committed record of a run made under an earlier version of the rule (Subsection 4.7).

Table E.21: Recourse evaluations. Plan: node: MVA/MWh. Condition: flexibility-price multiplier m (base: $m = 1$) or ageing calibration. Status: certified on the oscillatory (osc.) or monotone (mon.) branch, or the cause that left the evaluation uncertified (gap clause, lapse reset, growth test). k_0: first residual pass; k^*/end: certification cycle, or the cycle at which an uncertified evaluation stopped. Range: window range over τ. Band, and the magnitudes of the gap and slack, in k€. N.c.: non-clean cycles after k_0. Replay: last cycle replayed bitwise. Rule: applied in the run, or to the committed record. ^{f}: flagged certificate (a turning point at a non-clean cycle); ^{t}: bitwise twin of evaluation 2; ^{s}: superseded by evaluation 15. —: not applicable or not tabulated.

No.	Plan	Year	Condition	Status	k_0	k^*/end	Range/τ	Band	Gap	Slack	N.c.	Replay	Rule
(a) Baseline price: the plan without storage and single-node plans in 2025 (Tables 7 and 8, Fig. 5)													
1	$\boldsymbol{x} = \mathbf{0}$	—	base	cert.	—	181	0.93	4.2	—	—	—	—	record
2	7: 0.25/1	2025	base	cert.	—	172	0.96	4.4	—	—	—	—	record
3	7: 0.5/2	2025	base	cert., osc.	89	150	0.98	4.5	—	—	1	—	record
4	7: 0.75/3	2025	base	cert., osc.^{f}	88	142	0.72	3.3	—	—	1	88	run
5	7: 1/4	2025	base	cert., osc.	86	153	1.00	4.5	—	—	0	86	run
6	7: 1.25/5	2025	base	growth test	82	191	—	—	1.1	8.0	5	82	run
7	7: 0.5/1	2025	base	cert., osc.	104	175	0.86	3.9	—	—	0	104	run
8	7: 1/2	2025	base	lapse reset	94	203	—	—	0.5	8.1	2	94	run
9	7: 1.5/3	2025	base	growth test	89	198	—	—	0.8	13.8	6	89	run
10	7: 2/4	2025	base	cert., osc.	83	145	0.98	4.5	—	—	0	83	run
11	7: 2.5/5	2025	base	cert., osc.	83	138	0.99	4.5	—	—	0	83	run
12	5: 0.25/1	2025	base	cert., osc.	104	173	0.97	4.4	—	—	—	104	record
13	9: 0.25/1	2025	base	cert., osc.	103	173	0.92	4.2	—	—	—	103	record
14	9: 0.75/3	2025	base	cert., osc.	84	148	0.99	4.5	—	—	—	84	record
(b) Neighbours of the plan without storage (Tables 7 and E.22)													
15	5: 0.25/0.5	2025	base	cert., osc.	110	199	0.16	0.7	—	—	1	110	run
16	5: 0.25/0.5	2025	base	cert., osc.^{s}	110	167	0.99	4.5	—	—	—	110	record
17	7: 0.25/0.5	2025	base	cert., osc.	112	172	0.97	4.4	—	—	—	112	record
18	9: 0.25/0.5	2025	base	cert., osc.	113	174	0.81	3.7	—	—	—	113	record
19	5: 0.25/0.5	2030	base	cert., osc.	122	182	0.94	4.2	—	—	—	122	record
20	7: 0.25/0.5	2030	base	cert., osc.	125	195	0.99	4.5	—	—	—	125	record
21	9: 0.25/0.5	2030	base	cert., osc.	120	192	0.94	4.3	—	—	—	120	record
22	5, 9: 0.25/0.5 each	2025	base	cert., osc.	105	176	0.86	3.9	—	—	0	105	run
23	5, 7: 0.25/0.5 each	2030	base	cert., osc.	131	193	0.98	4.5	—	—	0	131	run
(c) Investment year (Table 14)													
24	7: 0.25/1	2030	base	cert., osc.	107	169	0.90	4.1	—	—	—	—	record

Table E.21 (continued)

No.	Plan	Year	Condition	Status	k_0	k^*/end	Range/τ	Band	Gap	Slack	N.c.	Replay	Rule
25	7: 0.25/1	2035	base	cert., osc.	106	167	0.89	4.1	—	—	—	—	record
26	7: 0.5/1	2030	base	cert., osc.	114	185	0.94	4.3	—	—	0	—	run
27	7: 0.5/1	2035	base	cert., osc.	112	174	0.84	3.8	—	—	0	—	run
28	7: 0.5/2	2030	base	cert., osc.	101	170	0.93	4.2	—	—	0	—	run
29	7: 0.5/2	2035	base	cert., osc.	103	172	0.85	3.8	—	—	0	—	run
(d) Flexibility price (Tables 9 and 10)													
30	$\boldsymbol{x} = \mathbf{0}$	—	$m = 1.5$	cert., osc.	93	153	0.91	4.1	—	—	0	93	run
31	7: 0.25/1	2025	$m = 1.5$	gap clause	85	194	—	—	2.5	6.3	0	85	run
32	$\boldsymbol{x} = \mathbf{0}$	—	$m = 1.75$	cert., osc.	108	139	0.79	3.6	—	—	—	—	run
33	7: 0.25/1	2025	$m = 1.75$	cert., osc.	101	121	0.65	2.9	—	—	—	—	run
34	$\boldsymbol{x} = \mathbf{0}$	—	$m = 2$	cert., osc.	125	146	0.48	2.2	—	—	0	125	run
35	7: 0.25/1	2025	$m = 2$	cert., osc.	123	150	0.90	4.1	—	—	0	123	run
36	7: 0.5/2	2025	$m = 2$	cert., mon.	135	198	0.94	4.3	—	—	0	135	run
37	7: 0.75/3	2025	$m = 2$	growth test	151	260	—	—	0.3	9.2	1	151	run
38	7: 1/4	2025	$m = 2$	cert., osc.[f]	127	158	0.71	3.2	—	—	1	127	run
39	7: 1.25/5	2025	$m = 2$	gap clause	159	270	—	—	8.8	6.2	1	159	run
(e) Twice the base flexibility price: the plan returned by the search and its neighbours (Subsection 4.2)													
40	5: 0.25/0.5; 7: 1/3.5	2030	$m = 2$	gap clause	—	281	—	—	9.2	1.5	—	172	record
41	5: 0.25/1; 7: 1/3	2030	$m = 2$	gap clause	—	261	—	—	9.3	1.0	—	152	record
42	5: 0.25/0.5; 7: 1/3	2030	$m = 2$	lapse reset	175	284	—	—	9.3	1.5	2	175	run
43	5: 0.25/0.5; 7: 0.75/3	2030	$m = 2$	cert., mon.	185	252	0.10	0.4	—	—	0	185	run
44	5: 0.25/0.5; 7: 1.25/3	2030	$m = 2$	cert., mon.	328	400	0.09	0.4	—	—	0	328	run
45	5: 0.25/0.5; 7: 0.75/3; 9: 0.25/0.5	2030	$m = 2$	gap clause	211	320	—	—	9.4	2.3	0	211	run
46	7: 1/3	2030	$m = 2$	cert., mon.	153	229	0.01	0.1	—	—	0	153	run
47	7: 0.75/3	2030	$m = 2$	cert., mon.	171	233	0.17	0.8	—	—	0	171	run
48	7: 1/3.5	2030	$m = 2$	cert., mon.	156	235	0.04	0.2	—	—	0	156	run
49	7: 1/4	2030	$m = 2$	cert., osc.	166	188	0.31	1.4	—	—	0	166	run
50	7: 1/3; 9: 0.25/0.5	2030	$m = 2$	gap clause	182	291	—	—	9.2	2.2	0	182	run
51	7: 1/3.5; 9: 0.25/0.5	2030	$m = 2$	gap clause	173	282	—	—	9.2	1.5	0	173	run
52	7: 0.75/3; 9: 0.25/0.5	2030	$m = 2$	cert., mon.	183	251	0.09	0.4	—	—	0	183	run
53	5: 0.25/0.5; 7: 0.75/2.5	2030	$m = 2$	cert., mon.	168	236	0.14	0.6	—	—	0	168	run
(f) Ageing calibrations of the reference unit (Table 11, Fig. 7)													
54	7: 0.25/1	2025	no ageing	cert., osc.	107	177	0.94	4.3	—	—	0	—	run
55	7: 0.25/1	2025	cycling only	cert., osc.	111	172	0.90	4.1	—	—	0	—	run
56	7: 0.25/1	2025	baseline	cert., osc.[t]	103	172	0.96	4.4	—	—	0	—	run

Table E.21 (continued)

No.	Plan	Year	Condition	Status	k_0	k^*/end	Range/τ	Band	Gap	Slack	N.c.	Replay	Rule
57	7: 0.25/1	2025	datasheet	cert., osc.	110	173	0.84	3.8	—	—	0	—	run
58	7: 0.25/1	2025	retention 50 %	cert., osc.	104	160	0.94	4.3	—	—	0	—	run
59	7: 0.25/1	2025	ret. 50 %, mid-block	cert., osc.	104	174	0.94	4.3	—	—	0	—	run
60	7: 0.25/1	2025	baseline, 0.50 floor	cert., osc.	112	184	0.92	4.2	—	—	—	—	run
61	7: 0.25/1	2025	baseline, 0.70 floor	cert., osc.^{t}	103	172	0.96	4.4	—	—	—	—	run

Note: Evaluations 1, 2, 40 and 41 continue earlier runs; their first residual passes are not tabulated (for evaluation 1, $k_0 = 123$, Fig. 9). Evaluation 16 is the first certificate of the 2025 node-5 neighbour; it rested on a cycle with a local solve that did not end optimal, and the neighbour was re-run (evaluation 15).

Appendix E.3. Neighbours of the Plan without Storage

The planning search evaluated all fourteen admissible unit neighbours of the plan without storage at its resolution σ_Q (Subsection 4.1). Table E.22 gives the eight of them that were re-evaluated under the certification rule: the smallest unit, 0.25 MVA / 0.5 MWh, at each node installed in 2025 or 2030, and at two pairs of nodes. Every one costs determinately more than the plan without storage.

Table E.22: Neighbours of the plan without storage re-evaluated under the certification rule: the smallest unit (0.25 MVA / 0.5 MWh) at the nodes listed. $\Delta = F(\boldsymbol{x}) - F(\mathbf{0})$, gross operational cost, settlement excluded; every difference is determinate.

Node(s)	Year	Eval.	Δ (k€)	Threshold (k€)	×
5	2025	15	49.7	12.6	3.94
7	2025	17	54.9	13.3	4.14
9	2025	18	52.5	12.6	4.15
5	2030	19	43.3	12.7	3.40
7	2030	20	49.0	13.5	3.62
9	2030	21	46.7	12.8	3.64
5 and 9	2025	22	102.1	12.6	8.09
5 and 7	2030	23	93.9	13.4	7.00

Note: The superseded first certificate of the 2025 node-5 neighbour (evaluation 16) gave 51.3 k€ (3.79×).

Appendix E.4. Static Arrangements: Solver Starts

Each static no-reverse-flow arrangement of Subsection 4.4 was solved from three starts (Table E.23); the best is reported in Table 12, and the spread over the starts is its band. The consistency pass is the second distribution solve at the transmission network's interface voltage (Section 3.6); the exceedance columns count the entries of that pass above the no-reverse-flow limit of the distribution networks and give the largest of them.

Appendix E.5. Evaluations Refused by the Gap Clause

Table E.24 lists the evaluations refused by the gap clause, with two others for comparison. At twice the base flexibility price the refused evaluations, and evaluation 42, whose rule was reset by two transmission solver recoveries before the gap clause was reached, share one fingerprint: a priced interface-consensus gap of −8.8 to −9.4 k€ at the cap, split about 55 %, 14 % and 31 % among nodes 5, 7 and 9, with the power-flow primal residual at 0.68–0.70 of its tolerance in the last cycle. The reference unit at 1.5 times the base

Table E.23: Static no-reverse-flow (NRF) arrangements at the plan without storage, by solver start. Cold: default start; perturbed: perturbed start; warm: started from the certified coordinated evaluation. Q: gross operational cost, settlement excluded. Curtailment: day-weighted, over the horizon.

Arrangement	Start	Q	NRF exceedances		Consistency	Curtailment (GWh)	
		(k€)	n	largest (p.u.)	pass (k€)	TN	DN
Coordinated	—	653 873.7	—	—	—	—	—
Passive NRF	cold	895 075.7	29	0.00618	−44.0	781.8	201.2
	perturbed[b]	895 049.9	29	0.00618	−69.8	781.7	201.2
	warm	895 049.9	29	0.00618	−69.8	781.8	201.2
Price-taker NRF	cold	744 770.3	70	0.000321	0.2	806.7	0.6
	perturbed	744 770.3	70	0.000321	0.2	807.3	0.6
	warm[b]	744 770.3	70	0.000321	0.2	807.6	0.6

Note: [b]: the start reported in Table 12; spread over the starts 25.8 k€ (passive) and 0.0 k€ (price-taker); reproducibility band of the coordinated evaluation 71.9 k€. Without any interface rule, from the cold start, the TN cannot accept the DN exchange in 1 of 12 blocks (6 h; 2035 Summer) under the passive arrangement and in 8 of 12 (25 h; 2025 Spring, 2025 Summer, 2030 Spring, 2030 Summer, 2030 Winter, 2035 Spring, 2035 Summer, 2035 Autumn) under the price-taker arrangement.

price is refused with a smaller gap (−2.5 k€), the same node split and a residual ratio of 0.027. The 1.25 MVA / 5 MWh plan at the base price is uncertified for a different reason, the growth test after repeated transmission solver recoveries, with a small gap of the opposite sign. The mechanism is described in Subsection 4.8.

Table E.24: Evaluations refused by the gap clause, and two others with the same or a contrasting signature. Gap: priced interface-consensus gap t_{sum} at the cap; share: its split among the interface nodes; residual: power-flow primal residual over its tolerance, last cycle; —: not recorded or none.

Eval.	Plan	Year	Cond.	Cause	Gap	Share (%)			Resid.	Lapse
					(k€)	n. 5	n. 7	n. 9		resets
31	7: 0.25/1	2025	$m = 1.5$	gap clause	−2.5	54.6	14.5	30.9	0.027	—
39	7: 1.25/5	2025	$m = 2$	gap clause	−8.8	55.2	13.7	31.0	0.682	160
50	7: 1/3; 9: 0.25/0.5	2030	$m = 2$	gap clause	−9.2	55.3	13.7	31.0	0.690	—
51	7: 1/3.5; 9: 0.25/0.5	2030	$m = 2$	gap clause	−9.2	55.3	13.7	31.0	0.690	—
45	5: 0.25/0.5; 7: 0.75/3; 9: 0.25/0.5	2030	$m = 2$	gap clause	−9.4	55.3	13.7	31.0	0.700	—
40	5: 0.25/0.5; 7: 1/3.5	2030	$m = 2$	gap clause	−9.2	—	—	—	—	—
41	5: 0.25/1; 7: 1/3	2030	$m = 2$	gap clause	−9.3	—	—	—	—	—
42	5: 0.25/0.5; 7: 1/3	2030	$m = 2$	lapse reset	−9.3	55.3	13.7	31.1	0.692	218, 222
6	7: 1.25/5	2025	base	growth test	1.1	58.3	11.3	30.3	0.012	—

Appendix E.6. Certification Statistics and Reproducibility

Certification statistics. The 42 single-scenario evaluations of the certification campaign are evaluations 3–11, 15, 22, 23, 26–31, 34–39 and 42–59 of Table E.21. Of these, 32 certified, 24 on the oscillatory branch and 8 on the monotone branch; the 10 uncertified divide into 5 refused by the gap clause, 2 reset by residual lapses and 3 failed by the growth test. The median certification cycle was 174, and certification came a median 65 cycles after the first residual pass (range 21–89); the median ratio of the window range to τ at certification was 0.86. The four evaluations added later (32, 33, 60 and 61) all certified. Ten of the 46 certificates the tables use stop within 5 % of τ (evaluations 2, 3, 5, 10, 11, 12, 14, 17, 20 and 23; evaluation 36, at 0.94 on the monotone branch, is just below), so the rule bounds the stopping error of each evaluation at about τ, not well inside it. The two evaluations continued past their certificates, 14 and 3, moved by a further 0.74τ and 0.78–0.88τ. After the first residual pass, 21 cycles in 10 evaluations had a local solve that was not clean, all in transmission blocks (column N.c.), while the 119 exits at IPOPT's acceptable level within ten times the tight-tail tolerances, all in distribution blocks, were counted as clean.

Reproducibility. Replaying a run reproduces its trajectory bitwise: the multi-scenario evaluation of the plan without storage replayed its 72 recorded cycles bitwise; the three single-scenario reference continuations replayed bitwise to their certification cycles; and every evaluation repeated from an earlier record replayed that record bitwise up to its first residual pass before continuing (column Replay). The evaluations ran one at a time on an Apple M2 Max with 32 GiB of memory under macOS 27.0, single-threaded: OMP, MKL, OpenBLAS, vecLib and NumExpr were capped at one thread in every evaluation's recorded environment, and the HSL library is built without OpenMP, so MA97 ran serially; the solver settings and versions are listed in Appendix A.8. Two distribution blocks (node 7 in 2025 Winter, node 5 in 2035 Winter) end some primary solves at IPOPT's acceptable level within ten times the tight-tail tolerances; they are counted as clean under the rule of Appendix A.5. Two offsets are common to every evaluation and cancel in every reported difference: tightening the interior-point tail lowered the certified objective of each of the three single-scenario references by 1.0–1.2×10^{-6} relative, with the same sign (the complementarity tolerance went to 10^{-6} from 10^{-4} in the distribution subproblems and from 5×10^{-4} in the transmission subproblem); and the 10^{-5} p.u. slack on the renewable availability bound lowers the objective by about 7.8 k€ at first order, 1.2×10^{-5} of it.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used ChatGPT in order to improve readability and language in selected parts of the manuscript. After using this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the published article.

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